

Explainability as a non-functional requirement in an AI-assisted veterinary clinical system: a case study

Authors

1. Alberto Jeanpool Marcillo Ponce^{1,*}
ORCID: 0009-0003-7886-3957
amarcillop@uteq.edu.ec
2. Robyn Willian Amagua Sacón¹
ORCID: 0009-0006-7363-4276
ramaguas@uteq.edu.ec
3. Anthony Alfredo Vera Gómez¹
ORCID: 0009-0007-8198-6199
averag10@uteq.edu.ec
4. Jhon Alexander Mesías Quijije¹
ORCID: 0009-0001-6929-7514
jmesiasq@uteq.edu.ec
5. Carlos Daniel Barrionuevo Fuentes¹
ORCID: 0009-0003-3415-3208
cbarrionuevof@uteq.edu.ec
6. Gleiston Ciceron Guerrero Ulloa¹ (academic supervisor)
ORCID: 0000-0001-5990-2357
gguerrero@uteq.edu.ec

¹Facultad de Ciencias de la Computación, Universidad Técnica Estatal de Quevedo (UTEQ), Quevedo, Ecuador.

*Corresponding author: Alberto Jeanpool Marcillo Ponce, amarcillop@uteq.edu.ec

Abstract

Context. AI-assisted diagnostic modules in veterinary clinical-management systems frequently lack explainability operationalized as a verifiable requirement.

Objective. To determine which explainability requirements, operationalized as verifiable non-functional requirements (NFRs), are necessary and sufficient for the different user profiles (technical and non-technical) of an AI-assisted veterinary clinical system.

Method. Applied single-case study grounded in Requirements Engineering. Requirements were elicited through semi-structured interviews and direct observation, analyzed qualitatively, operationalized as functional and non-functional requirements, and linked through requirements traceability.

Results. Open coding of 16 semi-structured interviews yielded 167 open codes, consolidated into 50 axial codes across 7 categories; closed-coding saturation was reached (1.33 new axial codes on average in the final three interviews, 2.67% of the cumulative total, below the 5% threshold). Acceptance of AI-assisted diagnostic suggestions was consistently conditional: participants required the veterinarian’s ability to review and correct suggestions (4/16) and grounding in validated literature and data-protection safeguards (6/16), with general moderate-to-high acceptance (11/16) and AI treated as decision support rather than a substitute for professional judgement. These conditions were operationalized as non-functional requirement RNF-18, traced to the AI-assisted diagnostic use case CU-06.

Conclusions. Explainability for AI-assisted veterinary diagnostic support is best operationalized around contestability and provenance of suggestions rather than technical model interpretability alone, and can be linked, through requirements traceability, to concrete verifiable requirements. Broader validation across additional cases and a second independent coding round remain as future work.

Note: this abstract must remain byte-for-byte identical to the abstract in the anonymized manuscript (`manuscripto_final.tex`). If either is edited, update both.

Keywords

Explainability; Non-functional requirements; AI-assisted diagnosis; Veterinary informatics; User studies

Declarations

Funding

The authors received no specific funding for this work.

Competing Interests

The authors declare no competing interests.

Ethics approval and consent to participate

Participation in the requirements-elicitation activities was voluntary and based on written informed consent. Participant information is handled using anonymized identifiers, and identifying materials are maintained separately in the restricted evidence area in accordance with the project’s data-protection procedures and Ecuador’s *Ley Orgánica de Protección de Datos Personales* (LOPD).

No individual ethics-committee approval identifier was issued for this project. This is consistent with the course-level ethical review process applied uniformly across all course teams for the Ingeniería de Requerimientos (ISR-401) Proyecto Fin de Curso at Universidad Técnica Estatal de Quevedo, rather than an omission specific to this study. The project

was classified as Category B (personal data, with an additional B1 requirement for host-organization endorsement) under this course-level framework. Institutional endorsement of the host-organization letter by the university’s Vicerrectorado Académico, required to formalize this classification, had not been issued as of the submission date; this affected all teams in the course uniformly and was not specific to this study.

Consent for publication

Participants provided written informed consent for the use of their responses for academic and research purposes. Only anonymized, aggregated findings are disseminated in this manuscript; no individually identifying information (names, signatures, or other identifiers) is published. Materials that could identify participants are retained separately in the restricted evidence area under encryption and are not intended for public dissemination (see Data, Materials and/or Code availability).

Data, Materials and/or Code availability

The research materials supporting the requirements-engineering phase include anonymized interview transcripts, thematic-coding materials, requirements artefacts, traceability information, and related project documentation. Potentially identifying materials, including original consent documents and multimedia evidence, are maintained separately in the restricted evidence area and are not intended for public dissemination.

The anonymized replication package is deposited on Zenodo: <https://doi.org/10.5281/zenodo.22558095> (version 2.0, CC BY 4.0). The protocol for the Enfoque 3 empirical component is registered on OSF: <https://osf.io/r5p8d/> (registered 1 August 2026).

Author contributions

Partial draft based on verified repository artefacts; must be reviewed and completed by all authors before submission, preferably following the CRediT taxonomy.

Marcillo Ponce, A.J.: thematic coding (open coding of interviews P09–P16; axial consolidation), thematic-saturation analysis, traceability-matrix verification, manuscript writing (original draft and revisions), project administration for this submission. **Amagua Sacón, R.W.:** traceability-matrix corrections and verification, open-coding saturation analysis script and figure. **Vera Gómez, A.A.:** Enfoque 3 empirical-component design, OSF protocol registration, validation-instrument design, confirmation of protocol deviations. **Mesías Quijije, J.A.:** independent second-coder inter-rater reliability analysis (Cohen’s κ), ethics-documentation package, ERS bibliography, walkthrough-validation evidence and minutes, member-checking session materials, organizational i* modeling scripts, field-notes reorganization. **Barrionuevo Fuentes, C.D.:** construction and maintenance of the statistical-analysis pipeline for the empirical component (data import, cleaning, descriptive statistics, assumption checks, hypothesis testing, effect-size estimation, figures and tables), reproducible data pipeline for the anonymized dataset, final resynchronization of results with the definitive sample (n=210). **Guerrero Ulloa, G.C.:** academic supervision.

All authors read and approved the final manuscript.

Use of AI-assisted technologies

During preparation and revision of this manuscript, an AI-assisted large language model (Claude, Anthropic) was used interactively, with human oversight, exclusively as a support tool for verification and language improvement. Its use was limited to: (i) checking the consistency, clarity, grammar, spelling, and wording of text previously prepared by the authors; (ii) verifying bibliographic and other factual data against the information and sources used by the authors; and (iii) suggesting improvements to the writing and overall clarity of the manuscript. The AI tool was not used to generate original or substantive content, including research data, thematic codes, categories, analysis, results, methodology, interpretations, conclusions, tables, figures, or research decisions. All suggestions were reviewed by the authors before inclusion. The authors remain fully responsible for the accuracy, provenance, and scientific integrity of all content and for the final version of the manuscript.

Acknowledgements

The authors thank the veterinary professionals, administrative personnel, and other participants who contributed to the requirements-elicitation and validation activities of the SGCV-IA project. Participant identities are not disclosed in this manuscript in accordance with the project's anonymization and data-protection procedures.